SOMONITOR: Combining Explainable AI & Large Language Models for Marketing Analytics

Aleksandr Farseev
SoMin.ai Research
Singapore, Singapore
sasha@somin.ai

Qi Yang
SoMin.ai Research
Singapore, Singapore
yang@somin.ai

Marlo Ongpin
SoMin.ai Research
Kuala Lumpur, Malaysia
marlo@somin.ai

Ilia Gossoudarev
ITMO University
Saint Petersburg, Russia
goss@itmo.ru

Yu-Yi Chu-Farseeva
SoMin.ai Research
Singapore, Singapore
joy@somin.ai

Sergey Nikolenko
ITMO University,
St. Petersburg Department of the Steklov Institute of Mathematics, Saint Petersburg, Russia
sergey@logic.pdmi.ras.ru

arXiv:2407.13117v2 [cs.CY] 10 Dec 2024

***Abstract*—Online marketing faces formidable challenges in managing and interpreting immense volumes of data necessary for competitor analysis, content research, and strategic branding. It is impossible to review hundreds to thousands of transient online content items by hand, and partial analysis often leads to suboptimal outcomes and poorly performing campaigns. We introduce an explainable AI framework *SOMONITOR* that aims to synergize human intuition with AI-based efficiency, helping marketers across all stages of the marketing funnel, from strategic planning to content creation and campaign execution. *SOMONITOR* incorporates a CTR prediction and ranking model for advertising content and uses large language models (LLMs) to process high-performing competitor content, identifying core content pillars such as target audiences, customer needs, and product features. These pillars are then organized into broader categories, including communication themes and targeted customer personas. By integrating these insights with data from the brand's own advertising campaigns, *SOMONITOR* constructs a narrative for addressing new customer personas and simultaneously generates detailed content briefs in the form of user stories that, as shown in the conducted case study, can be directly applied by marketing teams to streamline content production and campaign execution. The adoption of *SOMONITOR* in daily operations allows digital marketers to quickly parse through extensive datasets, offering actionable insights that significantly enhance campaign effectiveness and overall job satisfaction.**

***Index Terms*—Digital Advertising, Ads Performance Prediction, Deep Learning, Large Language Model, Explainable AI**

I. Introduction

The contemporary online marketing landscape is bursting with an overwhelming amount of content in the form of databases of advertisements, marketing campaigns of various competitors, properties of user audiences, and more. Marketers must not only sift through this information but also extract meaningful insights regarding audience segmentation, trending topics, and customer needs. Recent advances in the field of large language models (LLMs) have made them into tools of choice in various domains (see Section II). These models, trained on huge datasets, exhibit remarkable capabilities in both content analysis and content generation. In online marketing, modern LLMs are already able to offer a promising solution by processing and analyzing large datasets to identify key elements, facilitating a more targeted and effective marketing strategy. Moreover, they can help generate new advertisement content, both text-based and multimodal.

Despite all this progress, in this work we note that vanilla LLMs also have limitations in *content scoring*, which is a crucial aspect of effective marketing. We show that a small neural network, specifically tailored for CTR prediction based on advertising content, performs better at content scoring than standard LLMs, even those enhanced with multimodal capabilities, offering a more reliable tool for marketers.

Therefore, we propose a system, called *SOMONITOR*, that combines predictive analytics and LLMs in a unified online marketing assistant. Importantly, marketing strategies encompass more than just data analysis; they involve understanding and implementing effective communication styles. It is essential to utilize these communication styles not merely as classification labels but as dynamic personas. This approach enables the identification of different communication schemas tailored to diverse user personas. These schemas, and the user personas themselves, can be analyzed in existing marketing campaigns via a clustering procedure (Section IV). The next step involves aligning these personas with appropriate communication styles to craft compelling narratives. These narratives, or *stories*, are essential content that marketers can directly employ in their campaigns for brief production or content calendar building. However, to construct these stories one has to use high-quality data, including marketing communications and advertisements that have previously proven to be successful in building rapport with the corresponding user personas.

Thus, the core issue circles back to the need for an advanced, yet explainable, predictive model for content scoring and improvement, presented in Section VII. This model can be used to filter and identify "good" advertisements that can then serve to enhance the efficiency of LLMs in generating impactful marketing content. Our research addresses this need by introducing a predictive model that refines data input for LLMs, ensuring the generation of content that resonates

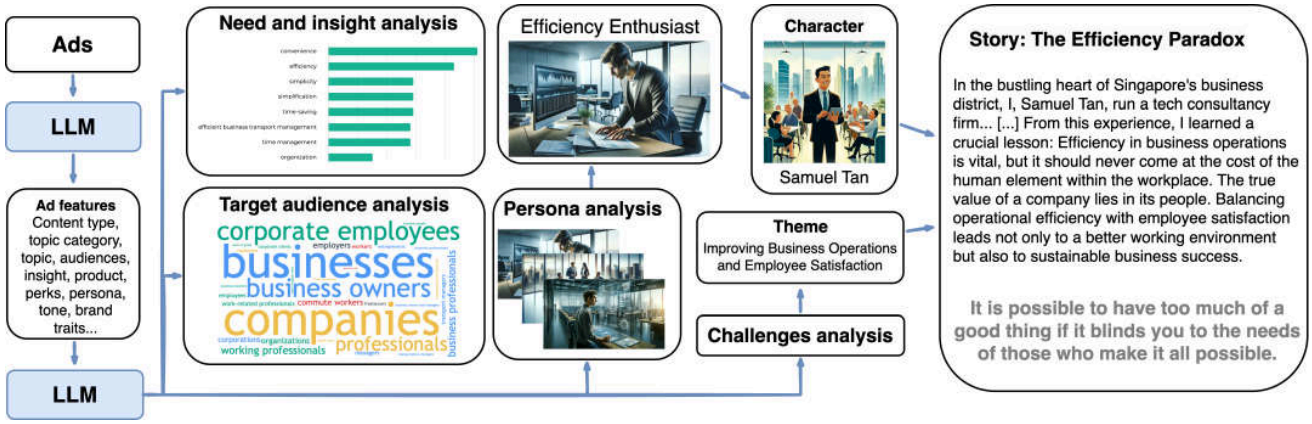


Fig. 1. General pipeline of the *SOMONITOR* advertising analysis framework.

well with target audiences and supports strategic marketing initiatives; in Section V, we show how this model alone can be used for iterative content improvement via interactive heatmap generation but also fits together with an LLM in a unified workflow that can generate complete user stories that being used by real businesses drastically improve their productivity. The overall pipeline of *SOMONITOR* is illustrated in Fig. 1.

II. Related work

Large language models (LLMs) have been successful in numerous applications [1], [2] and a wide variety of tasks, from low-resource machine translation [3] to writing source code for full-scale applications [4], [5] and solving problems that require mathematics and engineering design [6]. Importantly, the quadratic complexity bottleneck inherent in the original Transformer architecture [7] is also being overcome with sparse attention mechanisms [8], [9], low-rank decomposition of attention weight matrices [10], [11], Mamba-style models [12]–[14], and other modifications with linear complexity [15], [16]. As a result, latest LLMs can extend their context windows (i.e., input sizes) up to millions of tokens [17], [18], which makes it possible to use LLMs directly for mid- and large-scale analysis of text datasets.

Recently, LLMs have also obtained multimodal capabilities, allowing to use them for video understanding [19]. The recently released GPT-4o, where 'o' stands for "Omni", brings seamless multimodality of image, audio, and video analysis to state-of-the-art LLMs [20]. LLMs can be combined with diffusion-based generative models [21]–[23] to achieve state-of-the-art image and video generation capabilities [24]–[26].

Moreover, LLMs are increasingly used to process structured data, both in order to clean and/or augment data for ML model training and to directly improve the user experience. Extensive datasets used to train LLMs themselves [27]–[30] require rigorous filtering and ensuring data quality, for both technical [31]–[34] and legal considerations [35]–[37]. High-quality datasets are equally essential for LLM fine-tuning, particularly for instruction tuning, which depends on well-crafted full-text prompts and responses [38]–[40], and LLMs themselves increasingly assist in improving dataset quality; for instance, CoachLM [41] autonomously revises samples in instruction tuning datasets, IterClean [42] introduces an LLM-driven framework for cleaning data, and the Natural Instructions dataset [43] has been expanded with synthetic data in Unnatural Instructions [44]. Other efforts include purifying datasets of copyright and poisoning issues with smaller LLMs [45], among other approaches [46].

LLMs are also employed as tools for dataset preprocessing and refinement; e.g., Jellyfish [47] specializes in data preprocessing, LLMClean [48] focuses on cleaning tabular data, CAAFE [49] generates meaningful features for tabular datasets, and other works [50], [51] explore tabular data processing, even showcasing how LLMs can synthesize realistic synthetic data [52]. In the realm of databases and data lakes, LLMs have been applied to generate structured views [53], identify necessary data transformations [54], and enable natural language access to relational databases [55]–[59], e.g., for exploratory data analysis [60]. They also aid in interacting with graph databases [61], answering questions over relational datasets [62], and other related tasks [63], [64].

One important aspect of data processing with LLMs deals with recommender systems, where LLMs can unify tabular data processing and semantic understanding of items and context [65]. LLMs can be used directly for recommendation [66], [67], especially in conversational systems where the used is engaged in a dialogue with the system [68]–[70], but usually an LLM-based recommender is either trained fully with a separate Transformer-based architecture—such systems include the P5 approach (Pretrain, Personalized Prompt & Predict Paradigm) [71], [72], M6-Rec [73], and IRS [74]—or fine-tuned with instruction tuning [75], as in RecLLM [76], LC-Rec [77], TallREC [78], GLRec [79], or BERT-Trip [80]; text and Transformer-based embeddings of text can be used as user and item representations instead of ID embeddings [81], [82]. Multimodal LLMs have led to modality-based recommender systems that fuse encoders for different modalities [83], including sequence representations [84]; multimodality can make recommenders more robust to domain transfer [85]–[87].

Interpreting data through human-perceivable dynamic categories, such as topics or audiences, represents a significant step towards explainable AI. Nonetheless, the sheer volume of content and its prioritization continue to pose challenges, with limitations inherent in LLMs, in particular related to their hallucination phenomena [88]. We will see below that an LLM can say that the primary audience for 1000 pieces of marketing content is, e.g., "families", but without predictions of the content's popularity across this audience this insight is not actionable; LLMs are known to perform poorly and inconsistently in ranking problems [89]. At the same time, online content platforms such as social networks and search engines, namely *Facebook* (*Meta*) [90], *Google* [91], *Alibaba* [92], and *Taobao* [93], have invested considerable efforts towards content prioritization and scoring. Unfortunately, these results are based on private datasets, inaccessible to external content creators such as digital marketers. Proprietary engagement data used for content scoring models remains exclusive to the platforms, so independent creators cannot accurately predict or enhance the efficacy of their content.

From all the above, it is evident that the comprehensive capabilities of LLMs position them as high-level assistants across a myriad of tasks within the realm of online marketing. However, there exist relatively few research papers devoted to these potential applications [94]. The marketing industry is currently only starting to incorporate LLMs into the workflows, with many recent whitepapers and business surveys of possible applications, problems, and potential breadth of adoption of LLMs [95], [96]. Our recent work [97] presented a comprehensive online marketing analysis system based on large language models; in this paper, we build upon this work and combine and further improve LLM-based analysis with predictive analytics. A key advantage noted in [97] was the fact that LLM answers are understandable for humans and actionable in terms of business results: LLMs such as GPT-3.5 [98] and GPT-4 [99] were already able to explain the "reasoning" behind recommender models and provide aggregate insights about advertising campaigns. Before LLMs, text corpora in the context of recommender systems had been aggregated via topic modeling [100], [101], sometimes also with deep learning [102] and user profiling [103], [104], but topic modeling is based on the bag-of-words assumption and cannot summarize text as an LLM does. Visual understanding of ads has also been explored with convolutional networks [105].

However, a robust technology that enables effective differentiation of content sources for insight extraction, as well as a cohesive framework for data exploration that integrates both content differentiation and LLM-based explanations, is still sorely needed. Our work aims to address these gaps by developing methods that enhance the functionality of LLMs in online marketing applications, facilitating more efficient utilization of modern AI in real-world scenarios.

III. Content Pillar Identification

As noted above, data handlers in digital marketing frequently encounter large volumes of creative assets to analyze, along with numerous ongoing promotional activities from competitors in an "always-on" environment. Given this context, conducting detailed analyses for each advertisement or even for every specific competitor campaign becomes impractical, highlighting the need for automated analytical tools. These tools are crucial for assisting marketers in quickly assimilating, synthesizing, and driving inferences from extensive data and information available to them.

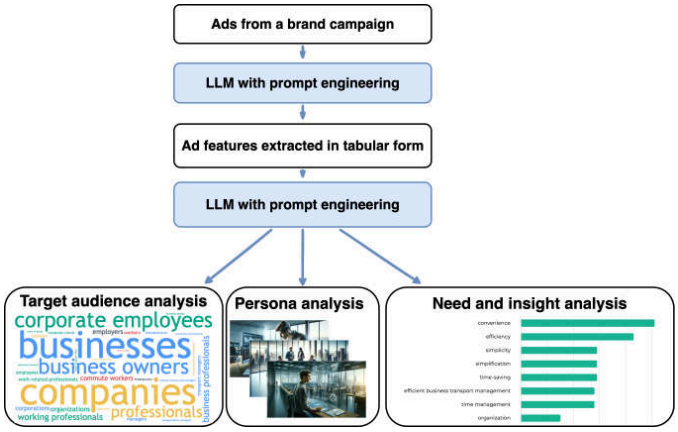


Fig. 2. General pipeline of our LLM-based analysis

To address this challenge, we propose to integrate a novel method into our framework that leverages LLMs to provide semantic insights into content, referred to as "content pillars" below. We outline an analytical pipeline that incorporates explanations and generations based on LLMs, demonstrating its practical utility through an empirical scenario involving four Singapore-based telecommunication companies. This component of our framework aims to enhance the interpretability and understanding of data, thereby supporting more informed decision-making in these dynamic and competitive sectors.

The architecture of our analytics is illustrated in Fig. 2. Initially, we employ an LLM to extract specific, well-defined insights from the input content, such as the customer needs addressed by the advertisement and the products being advertised. These insights are then archived as *content pillars* in a tabular format. Next, these pillars are processed with custom-crafted prompts to perform a generalized analysis of a brand's target demographics, personas, needs, and insights conveyed by the advertisements, as well as the tone and topical categories of both current and previous campaigns. This comprehensive analysis mirrors the campaign assessments typically conducted by marketing professionals and can be further used to refine a brand's messaging, tone, and target demographics. Below we explore these elements in detail.

Figure 3 presents sample qualitative results of our experiments on the analysis of content pieces. We curated batches of advertisements from the *Facebook Ad Library* for the same brand and processed them through an LLM with custom prompts. As a result, the LLM successfully identified key attributes of each advertisement, providing adequate responses to ostensibly "human" inquiries such as pinpointing the human



Fig. 3. Sample ad analysis

need, insight, and predominant archetypes in a specific ad. Note that responses to most queries are standardized (as directed by the prompt) and amenable to automated processing. This analysis has always been important in online marketing but to our knowledge has not been automated and scaled before, always requiring human labor and thus being limited to a small sample of ads rather than an extensive dataset.

IV. Mining Communication Personas and Themes

A. Problem setting

In the field of marketing, utilizing explainable data descriptors such as the content pillars is essential. However, to significantly enhance a brand's marketing and communication strategies it is necessary to go beyond mere description and engage in inferential data analysis. Unfortunately, marketers often lack the necessary skills in data operations, hindering their ability to derive meaningful conclusions from the data, thereby requiring additional support. To bridge this gap, we propose a unified framework *SOMONITOR* that unites LLM-based analysis with content scoring, designed to provide marketers with higher-level content observations.

In marketing, the narrative constructed by any content emerges from an underlying story, which is a synthesis of a communication theme and a specific customer persona that the campaign aims to engage. We will come back to the stories in Section V, illustrating this concept through a representative example, highlighting the integral relationship between the storytelling elements and the targeted persona. This approach empowers marketers to craft more resonant and effective communication strategies by aligning their messaging with the expectations and needs of their intended audiences. But to get to a good story, we need to extract two essential storytelling components: personas and challenges.

B. Persona Identification

In marketing, the *customer persona* is developed based on data about the audience of a brand, supported by other content pillars such as customer needs, interests, and aspirations [106]. We have used X-Means clustering [107] to divide the content pieces into distinct customer personas based on brand content, specifically clustering ADA embeddings [108] of the "Audience" content pillar as the main semantic differentiator. X-Means was run with 3 starting centroids and $k_{\max} = 50$, identifying the number of clusters with BIC values [107].

For this study, we analyzed data from the Gojek car-hailing brand and its main competitor in Singapore, *Grab*, focusing exclusively on the B2B business audience to limit the number of personas. The data for this experiment, collected over 2023 and 2024, comprised 5,967 content pieces, with 3,703 published as ads and 2,264 as organic social media posts. Since most of the content related to B2B product offerings for both brands was presented in ads, we focused solely on the advertising content, totaling 1,120 ads related to business products. Brand-wise, *Gojek* ran the majority of the ads (76.8%, or 849 ads), while *Grab* accounted for the remaining 23.2% (271 ads).

Next, we annotated the clusters, generating names and descriptions by prompting an LLM to do so based on the predominant parameters and content of the posts associated with each cluster. Finally, an image for each persona was generated by the LLM in accordance with the persona's description. Fig. 4 shows the resulting three distinct clusters, each representing a unique customer persona along with their associated characteristics: "Efficiency Enthusiasts" (Persona 1), "Financial Empowerment Champions" (Persona 2), and "Efficiency Innovators" (Persona 3), with content counts of 206, 144, and 707 respectively. Note that some content items did not correspond to any of the three identified personas (i.e., represented outliers) and were thus excluded from the final tally, explaining the disparity with the initial 1,120 ads.

As a result, notable distinctions between the personas have emerged. Both Persona 1 and Persona 3 focus on organizational efficiency, albeit through different approaches: Persona 1 advocates for efficiency through traditional methodologies, while Persona 3 champions it through innovative strategies. In contrast, Persona 2 stands apart by primarily emphasizing the need for financial empowerment of employees.

C. Communication Theme: Challenges

A persona is an essential but not sufficient component needed to come up with a marketing story and content briefs. In particular, it is important to identify a communication scheme to be combined with the persona's key attributes such as background or character, so that the story would get proper context and would grab attention [106]. To explore this, we have performed X-Means clustering based on the "Insights" content pillar, again using ADA embeddings for text [108] and running X-Means with 3 starting centroids and $k_{\max} = 50$.

Similarly to persona clustering, we annotated the resulting challenge clusters and generated parameters such as cluster

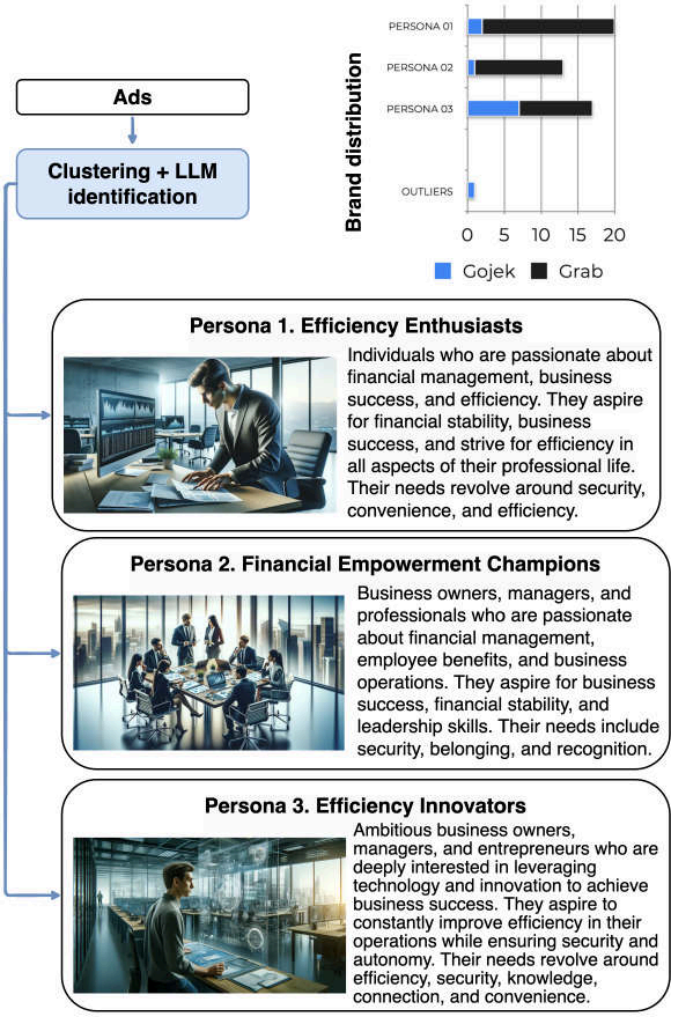


Fig. 4. Sample persona analysis results

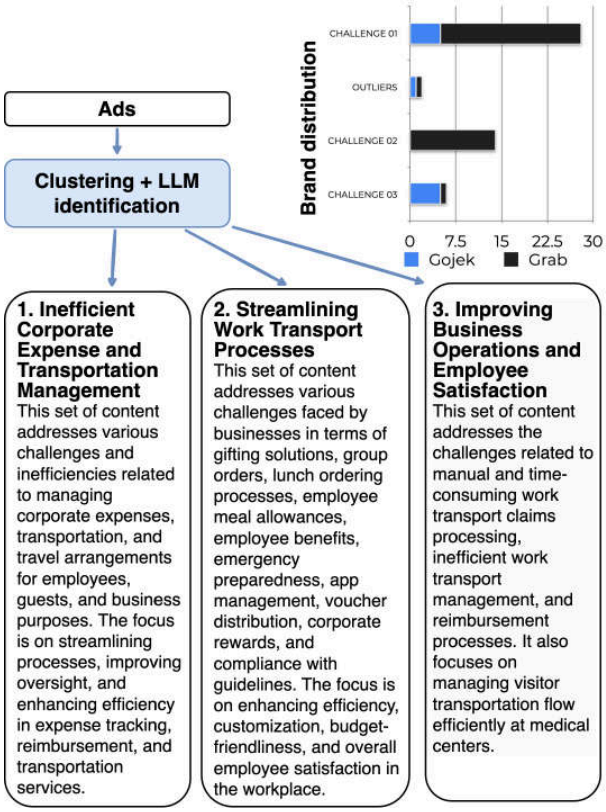


Fig. 5. Sample challenges mined by our system

name and cluster description by prompting an LLM to do so based on predominant parameters and content of the posts associated with each cluster. Figure 5 shows sample results: in this case, we have identified three distinct clusters, each representing a unique customer challenge along with their associated characteristics: "Inefficient Corporate Expense and Transportation Management" (Challenge 1), "Improving Business Operations and Employee Satisfaction" (Challenge 2), and "Streamlining Work Transport Processes" (Challenge 3), with content counts of 672, 94, and 336 respectively.

V. Data-Driven Story Generation

In order to establish a data-driven narrative that would be useful, easy to understand, and attractive to content marketing professionals, it is essential to delineate the content and marketing strategies a brand may deploy, guided by the analysis of the best-performing content from competitive brands. At this point, we have already discussed how the *SOMONITOR* framework extracts personas and challenges based on advertising content prefiltered by a CTR prediction and ranking model. Now it is time to combine them into a unified marketing story.

Consider the scenario where we are part of a marketing team at *Gojek*, tasked with identifying a suitable communication theme and a target persona for a campaign aimed at market expansion through new audience engagement. It is desirable to opt for a persona and theme that are presently underexploited, thereby expanding our audience reach. As illustrated in Fig. 4, the analysis shows that Persona 1, labeled "Efficiency Enthusiasts", is more heavily leveraged by *Grab* than by *Gojek*, indicating a potential niche opportunity for *Gojek*. At the same time, as shown in Fig. 5, the communication theme "Challenge 3: Streamlining Work Transport Processes" is more frequently addressed by our competitor, *Grab*, but is underrepresented in *Gojek*'s content. This observation advocates for adopting this theme for persona-centric narrative generation.

Consequently, adhering to the framework outlined in Figure 6, we prompt an LLM to amalgamate the characteristics of the selected persona and the chosen communication theme to devise a narrative for a fictitious character, "Samuel Tan", a business owner in Singapore. The pipeline for this generation is shown in Figure 6: character description is also generated by a custom-prompted LLM, and then the character and theme are combined to create a compelling story. The story is abbreviated in Fig. 6, in reality it explains over several paragraphs how Samuel prioritizes business efficiency but often neglects employee satisfaction, leading to increased

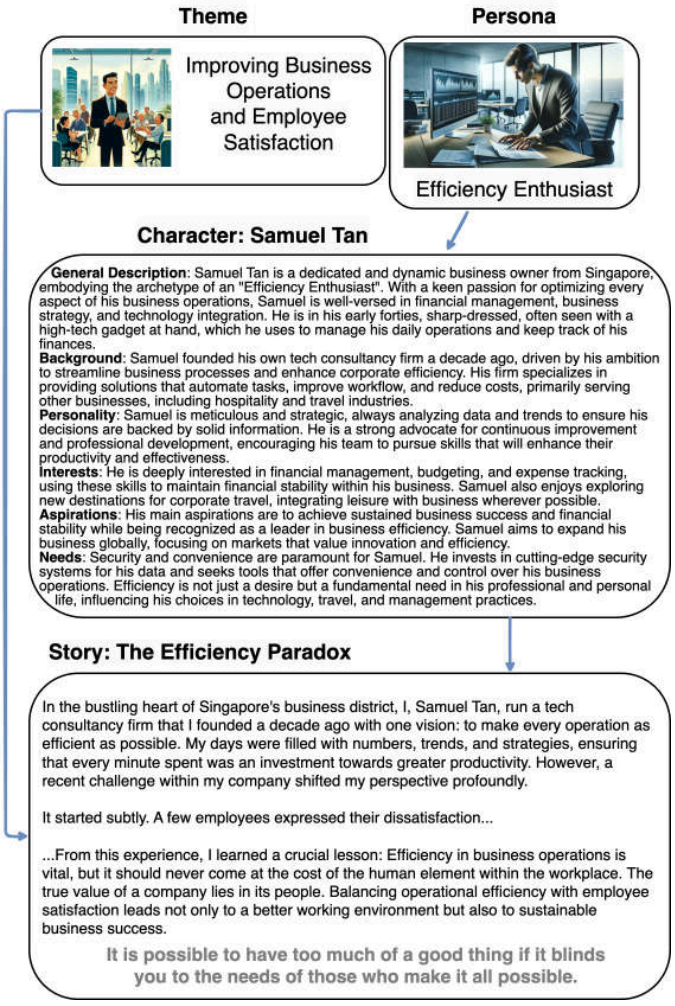


Fig. 6. User story generation: pipeline and an example.

turnover rates. Recognizing this issue, Samuel opts for *Gojek* not merely as an efficiency solution but as a means to reengage with employees, enhance their efficiency, and improve overall job satisfaction; the story ends with a conclusion (emphasized in Fig. 6) that shows an important insight that Samuel has obtained by engaging with *Gojek*. This narrative could serve as a content brief for a creative agency or be utilized as a pitch for a novel marketing campaign, previously unexplored by the marketing team in such depth and with such rapid production capability. In our professional opinion, content briefs produced by *SOMONITOR* are on par with similar human-written narratives and can be used just as effectively.

VI. Case Study: Data-Driven Story Generation

This section presents a case study evaluating the impact of the proposed story generation approach on marketing workflows across four organizations: NEO360, Blak Labs, Bit by Bit Marketing, and Mothercare. These organizations implemented the system, documented the time required for routine tasks before and after adopting the story generation approach, and reported substantial reductions in time spent on key tasks such as content strategy, campaign planning, competitor monitoring, and audience research. This evaluation was conducted across multiple brands within agency settings and for various product categories at Mothercare.

TABLE I
Time Savings with Semantic Enrichment Approach

Task	Before (Hours)	After (Hours)
NEO360 (Performance Agency)		
Social Media Proposal Preparation	3.0	0.75
Competitor Monitoring	3.0	0.25
Audience Research	5.0	0.50
Brainstorming Ad Angles	2.5	0.25
Blak Labs (Creative Agency)		
Campaign Planning	5.0	2.0
Content Planning	10.0	3.0
Brand Analysis	10.0	1.0
Bit by Bit Marketing (Branding Agency)		
Content Strategy	5.0	1.0
Mothercare (Consumer Brand)		
Competitor Analysis	5.0	2.0
Ad Compilation	5.0	1.0
Customer Segmentation	5.0	2.0

1) NEO360: Streamlining Performance Marketing: NEO360[1], a performance agency, implemented the semantic enrichment approach to enhance workflows for tasks such as social media preparation, competitor monitoring, audience research, and brainstorming ad angles. As summarized in Table I, the system reduced task durations by 2x to 3x; e.g., the time required for social media proposal preparation was reduced from 3 hours to 45 minutes, while competitor monitoring was shortened from 3 hours to just 15 minutes, showing radical acceleration of research-intensive tasks.

2) Blak Labs: Optimizing Creative Workflows: Blak Labs[2], a creative agency, leveraged the system to streamline campaign planning, content planning, and brand analysis. The results, shown in Table I, demonstrate time reductions of 2x to 3x. Notable gains include reducing the time for brand analysis from 10 hours to 1 hour and content planning from 10 hours to 3 hours, proving that the platform can handle resource-intensive creative processes.

3) Bit by Bit Marketing: Enhancing Branding Strategies: Bit by Bit Marketing[3], a branding agency, used the semantic enrichment approach to optimize its content strategy workflow. Table I shows that the agency achieved a 5x reduction in time spent on this task, from 5 hours to 1 hour; in this case, the system was able to streamline strategic processes, enabling agencies to focus on higher-level planning and execution.

4) Mothercare: Improving Consumer Brand Efficiency: Mothercare[4], a consumer brand, applied the system to optimize

[1]https://neo360.digital/
[2]https://blaklabs.com/
[3]https://bitbybitmarket.com/
[4]https://www.mothercare.com.sg

TABLE II
EXPERIMENTAL RESULTS FOR *SOMONITOR*.

Brand	nDCG@5	nDCG@10	Recall@3	Recall@5
A	0.498	0.796	0.333	0.4
B	0.541	0.821	0.333	0.4
C	0.588	0.829	0.6	0.6

TABLE III
PERFORMANCE EVALUATION AGAINST LLM BASELINES; "GD" REPRESENTS A GROUNDED MODEL.

	nDCG@5	nDCG@10	Recall@3	Recall@5
GPT4-gd	0.591	0.818	0.333	0.6
GPT4	0.465	0.755	0	0.4
GPT4o-gd	**0.599**	0.827	0.333	0.6
GPT4o	0.511	0.750	0.333	0.4
Gemini-gd	0.578	0.817	0.333	0.6
Gemini	0.564	0.769	0	0.6
SOMONITOR	0.588	**0.829**	**0.6**	0.6

competitor analysis, ad compilation, and customer segmentation workflows. Table I highlights the 3x to 5x improvements achieved; e.g., ad compilation time was reduced from 5 hours to 1h, and customer segmentation tasks were completed in less than half the time. These results validate our approach's versatility in enhancing productivity across diverse domains.

Results of this case study highlight the transformative impact of data-driven story generation in marketing workflows. Our LLM-based system not only reduces task durations but also provides actionable insights for optimizing operations. The proposed platform can address real-world challenges and enable marketers to achieve huge productivity gains while maintaining high-quality outcomes. This evidence strongly supports the adoption of semantic enrichment and data-driven content optimization in marketing practices.

VII. CONTENT SCORING

A. Problem setting and dataset

To make actionable recommendations, one has to enable marketers to evaluate data and select which content to incorporate into analytics, ideally based on anticipated performance. This leads to the need for content scoring, so in this section we add to *SOMONITOR* a system for the scoring of advertising content, developed in our previous work [97], and evaluate it against state of the art LLMs to prove its continued relevance.

To assess the effectiveness of our recommender system, we conducted a validation study using real ad creatives and their performance data. We collected real-world ad creatives and their performance data from three brands on *Facebook*, each with distinct marketing objectives: sales, conversion, and traffic, spanning from October 1, 2023 to December 31, 2023. The dataset comprised text and images of their ads alongside their performance measured as CTR values. We chose three (anonymized) brands: Brand A (16 ads in the fitness industry), Brand B (29 ads in the education industry), and Brand C (16 ads in the automobile industry).

The ads were assessed not just according to the brand affiliation, but also according to the Ad objective selected on the Meta Ads platform, which is a crucial component of evaluation in our scenario due to the fundamental differences in click-through rates across various industries and marketing goals that Ads achieve with different Ad objective settings [97]. For instance, advertisements optimized for clicks may result in lower costs per click and higher click-through rates, yet they may not effectively convert customers. Conversely, ads aimed at achieving better conversion rates might not exhibit as high click-through rates as those optimized merely for clicks.

B. Content scoring models

We use our previously developed content scoring approach [97]; it was originally designed to predict CTR by classifying ads into CTR performance categories but here we operate in a recommendation setting, i.e., pose content scoring as a ranking problem because (i) marketers do not need precise CTR values to choose the best content but rather a comparative assessment, (ii) CTR values undergo significant distribution shift across platforms, time, and other external factors, but the ranking of ads should be more stable, and (iii) the LLM performance in qualitative ranking should be significantly better than in numerical predictions.

Thus, we take the SODA model from [97] that predicts a distribution over "High CTR", "Average CTR", and "Low CTR" labels and add a linear layer that translates the probabilities into a single score via a trainable linear combination:

$$\text{Score} = \alpha \cdot p(\text{HighCTR}) + \beta \cdot (1 - p(\text{HighCTR})),$$

converting a classification model into a ranking model.

On the side of LLMs, we used GPT-4 [99], its latest multimodal variant GPT-4o, and Gemini 1.5 Pro [17] for comparison with *SOMONITOR*. A significant challenge in utilizing LLMs for this task is their tendency to hallucinate, which can result in generating inaccurate or nonsensical information that compromises the reliability of their output. To mitigate this issue and ground the LLM-based approach to ensure the predicted data was trustworthy, we implemented several measures to reduce randomness: (i) *temperature adjustment*, setting the temperature parameter of the LLMs to 0.1, which is known to improve stability by reducing the variability in the model's outputs; (ii) *grounding*, i.e., including another three sample ads from the brand within the prompt, each with a specified CTR rank: one with the best performance, one with average performance, and one with the worst performance; (iii) *ensembling*, running the ranking process with LLMs five times in the same settings and ranking the ads by the (lowest to highest) sums of their ranks from all runs.

C. Results

For evaluation, we use standard ranking-based metrics, namely Recall@3 and Recall@5. Recall is very important in practice, especially in resource-constrained domains such as online advertising campaigns, where top k recommendations should encompass a substantial portion of the truly relevant

items, so that marketers can focus their efforts on the most promising opportunities. We have also computed normalized discounted cumulative gain (nDCG) at cutoffs 5 and 10, another commonly used way to assess ranking quality. Table II shows a case study across three major industries. In the fitness sector, *SOMONITOR* achieved moderate effectiveness with nDCG@5 of 0.498 and nDCG@10 of 0.796, alongside Recall@3 and Recall@5 scores of 0.333 and 0.4, respectively. Brand B, representing education, showed improved performance, with nDCG@5 of 0.541 and nDCG@10 of 0.821, but recall stayed the same. In the automobile industry results improved, achieving nDCG@5 of 0.588 and nDCG@10 of 0.829, with recall@3 and recall@5 both scoring 0.6. These findings show that content scoring quality varies across industries and highlight the importance of industry-specific optimization to enhance recommendation accuracy.

Then we evaluated *SOMONITOR* against LLM baselines. In campaigns focused on driving traffic, the primary goal is to maximize click-through rates (CTR), which serve as a measure of user engagement and interaction with the advertisement, which is particularly pertinent given the distinctive challenges posed by traffic-driven campaigns as opposed to conversion-oriented strategies. Thus, we evaluate the performance with the data from Brand C, a traffic-oriented campaign. Table III presents the results from a case study focused on Brand C, comparing the performance of *SOMONITOR* against modern LLMs. It shows a significant performance boost achieved by knowledge grounding across various models and metrics; e.g., nDCG@5 for GPT-4 improved from 0.465 without grounding to 0.591 with grounding, while its Recall@5 increased from 0.4 to 0.6. This also holds for GPT-4o and Gemini1.5 Pro, showing that grounding is critical in enhancing the relevance and quality of recommendations. Ungrounded LLMs exhibit markedly lower performance, particularly in the Recall@3 metric: both Gemini and GPT-4 failed to place any high-performance items within their top 3 results at all.

As a result, the prediction model outperforms LLMs in recommending relevant items, achieving the highest nDCG@10 of 0.829 and recall@3 of 0.6 among compared models. *SOMONITOR* is better able to balance ranking accuracy and item recall, delivering highly relevant and stable (especially compared to individual LLM results) recommendations to users. Notably, these outcomes are achieved without any fine-tuning (grounding) using data from the actual industry-specific distributions: we used a pretrained model. This offers significant potential for further improvement of results should such fine-tuning be implemented, a direction we intend to explore in our future work. Improvements in LLMs' Recall@k due to grounding also hint at the critical role of domain knowledge and context in enhancing the relevance and quality of recommendations. Ungrounded LLMs, relying solely on their pre-trained language models, struggle to capture the nuances and specifics of the recommendation task, resulting in sub-optimal item retrieval. By incorporating domain-specific knowledge and grounding the models in the relevant context, the recommendations become more tailored and relevant.

We note that our proposed *SOMONITOR* framework outperforms the baselines in both Recall@3 and Recall@5, achieving a Recall@3 score of 0.6 which is higher than any LLM's result. This supports our conclusion that the online marketing industry still cannot fully rely on LLMs for content scoring and explains our decision to include a CTR prediction and ranking model into the mostly LLM-based *SOMONITOR* framework.

VIII. Quantitative Evaluation and Case Study

We have shown that a small neural network can outperform one-shot fine-tuned LLMs in predicting content performance prior to the official release of advertisements. However, for this solution to be practical for marketers it must not only identify what works better but also provide actionable guidance on how to improve the content. We have noted before [97] that attention layers of the content scoring network can be visualized to assist marketers in refining their content iteratively, but have not tested it in real world conditions until now. In this section, we evaluate this approach with the *Meta* advertising engine, aiming (1) to assess whether regions highlighted by attention heatmaps correlate with higher user engagement and (2) to determine whether removing key objects identified by heatmaps adversely affects campaign performance.

A. Experimental Setup and Methodology

We used a diverse dataset of advertising creatives with multiple visual elements and processed them through our predictive framework, which generated attention heatmaps to identify areas likely to maximize user engagement.

Selection of Creative Assets: only creatives with high-confidence predictions (above the model's average confidence threshold) were included in the experiment. This ensured that selected assets featured impactful elements, allowing for a focused assessment of the heatmap's predictive capabilities.

We used the following experimental workflow:

1) *heatmap generation:* each creative was processed with *SoWide-v2*, to generate heatmaps (Fig. 7), highlighting elements expected to drive user engagement;
2) *iterative object removal:* based on the heatmaps, critical objects were sequentially removed, and updated heatmaps were generated after each iteration; this process produced three modified versions of each original creative;
3) *campaign deployment:* original and modified creatives were deployed in a real-world campaign on the *Meta* platform with identical settings (*Objective*: Landing Page View; *Placements*: All; *Budget Control*: automatic; *Ad Copy*: "Unlock Game-Changing Marketing Insights in Seconds with SOMONITOR"; *Duration*: 1 day);
4) *performance metrics:* key performance indicators such as click-through rate (CTR), click-to-landing page view (CTR-LPV), and landing page views (LPV) were monitored for all creatives, with a total of 65,107 impressions, 2,243 clicks, and 1,162 landing page loads recorded;
5) *performance analysis:* to quantify the impact of object removal, we counted the proportion of cases where performance declined after object removal.

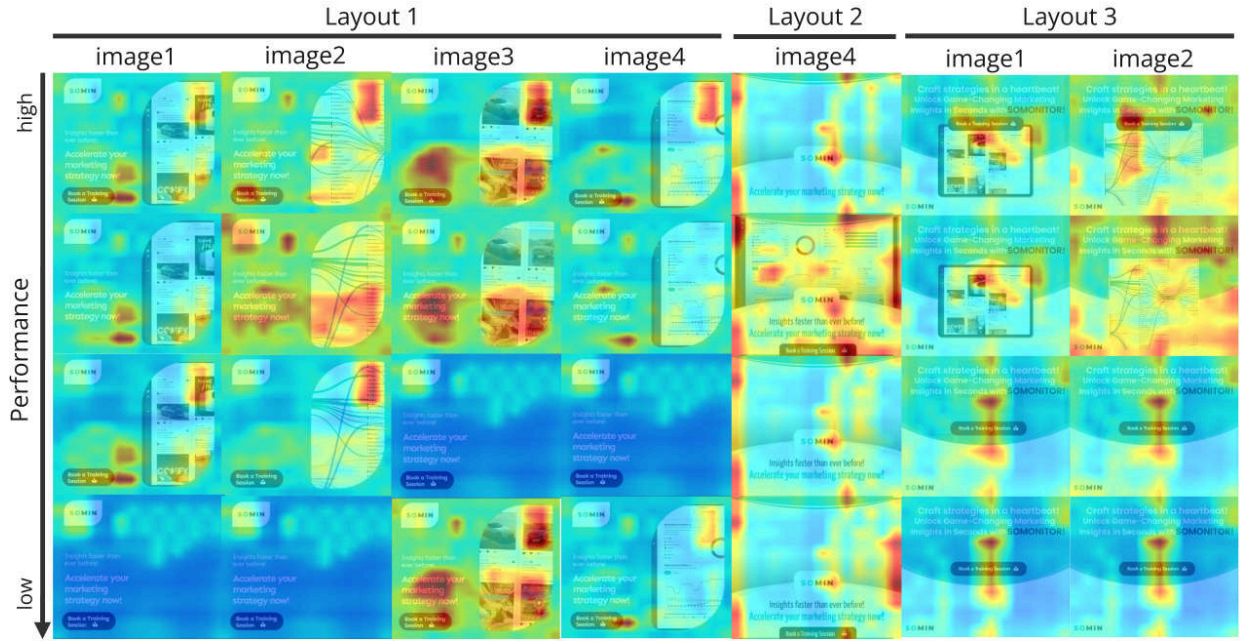


Fig. 7. Visualization of object removal guided by attention heatmaps.

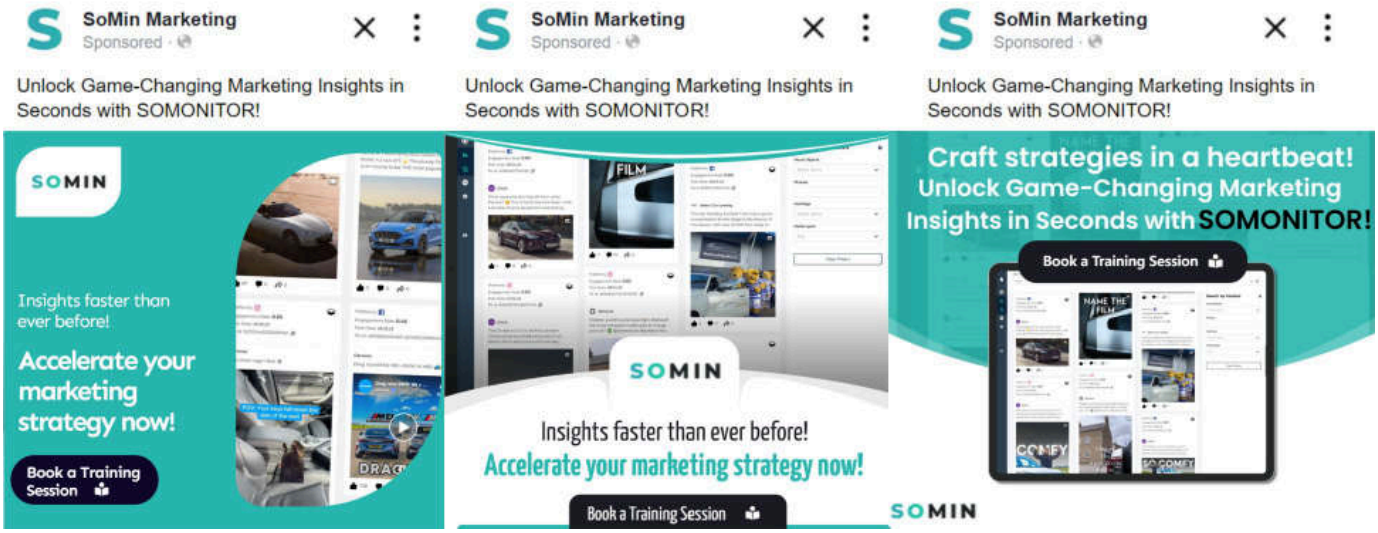


Fig. 8. Example of an ad variant deployed on the Meta platform.

TABLE IV
EVALUATING OBJECT REMOVAL IN A REAL-WORLD CAMPAIGN

Layout	LPV	CTR-LPV	CTR	F1 Score
Layout 1	0.769	0.769	0.692	0.750
Layout 2	0.751	0.751	0.250	0.500
Layout 3	0.570	0.857	0.857	0.860
Overall	0.708	0.792	0.667	0.740

B. Results and Observations

Experimental results summarized in Table IV show the efficacy of heatmaps for guiding ad optimization:

- *Layout 1:* Moderate performance with well-aligned LPV and CTR-LPV values (0.769), indicating consistent user engagement. The F1 score (0.750) suggests accurate predictions of ad performance.
- *Layout 2:* Significantly lower CTR (0.250) shows problems in attracting initial clicks despite robust post-click engagement (CTR-LPV 0.751), reflecting the model's limitations in detecting less impactful visual cues.
- *Layout 3:* The highest CTR-LPV (0.857) and F1 score (0.860) demonstrate strong alignment between predicted and actual user engagement. However, lower LPV (0.570) indicates potential gaps in predicting post-click retention.

C. Impact of Object Removal

Systematic removal of key visual elements led to the following results: (i) removing the *CTA Button* caused a 22% drop in CTR, showing its importance in driving engagement. (ii) removing *transparent background* led to the most severe performance decline, with a 64% drop in CTR, highlighting its influence in guiding user attention; (iii) removing the *primary visual element* resulted in a 58% CTR drop, demonstrating its significance in capturing initial user attention. The model's predictions are closely aligned with observed performance.

IX. Gen-AI Performance Marketing Consultant

We have been exploring strategies that enable marketers to enhance their content efforts, as well as prevalent trends in content marketing. However, major companies like *Procter & Gamble* or *McDonald's* often undergo rigorous reviews before altering creative materials or ad copy. This can make proposed improvements to advertising content impractical, as the targeted campaign may end before suggestions are implemented. When creative changes are unfeasible, real-time campaign optimization becomes essential. Even for campaigns with high-quality creatives, this approach ensures precise alignment of creatives with target audiences and helps identify performance bottlenecks through detailed data analysis. In this section, we discuss how LLM-based analysis can help evaluate key performance metrics and guide strategic decisions in campaign optimization, connecting the micro-level analysis of ad creatives with the macro-level approach of the LLM helping steer the course of the entire advertising campaign.

A. Metrics for Measuring Campaign Performance

We expand the focus from ad creative analysis to a holistic view of campaign performance using several key metrics:

- *Reach*: total unique individuals exposed to the campaign, indicating its audience breadth and potential impact on brand visibility and awareness.
- *Frequency*: average number of times a user sees the ad; tracking it is needed to avoid audience fatigue [109];
- *Results*: total targeted conversions (e.g., leads or purchases), the primary measure of campaign success;
- *Cost per Result*: average cost per conversion, reflecting campaign efficiency and guiding budget optimization;
- *Amount Spent*: total expenditure, needed for budget tracking and ensuring alignment with campaign goals;
- *CPM* (Cost per 1,000 Impressions): cost of reaching 1,000 users informs about market competition [110];
- *CTR* (Click-Through Rate), the prediction target for our *SoWide-v2* model (Section VII); it assesses ad engagement and content effectiveness;
- *CR for Link Clicks to LP Views*: fraction of users who click the ad and view the landing page, indicating landing page effectiveness;
- *CR for Link Clicks to Results*: fraction of users clicking the ad who achieve the targeted conversion, directly reflecting campaign goal attainment.

These metrics collectively provide a comprehensive view of campaign performance, encompassing audience engagement, conversion rates, budget efficiency, and ad quality, enabling informed optimization decisions.

B. Data Acquisition and Preprocessing

Metrics data for the *Meta* campaign were collected over a three-month period (October 1, 2023 – January 23, 2024) at three levels of granularity: (i) *weekly*, which provides a consolidated view by smoothing short-term fluctuations (Fig. 10); (ii) *daily*, which offers detailed insights into day-to-day variations essential for detecting anomalies or immediate external influences, and at the (iii) *creative level*, which evaluates the contribution of individual creatives to campaign performance. To adapt data for LLM analysis, it was preprocessed as follows: (i) *creative data* represented as tabulated insights with images resized to 512×512 pixels and encoded in base64; (ii) *numerical metrics*, as percentage change rates between consecutive data points to focus on trends and fluctuations rather than raw values. This preprocessing ensures that metrics, diverse in granularity and spanning over 115 days, are strategically well-structured for comprehensive campaign analysis.

C. Prompt Engineering

Automated campaign performance analysis requires a well-crafted prompt in order to guide the LLM through a comprehensive evaluation of key metrics and their interdependencies. Our prompt format follows the convention of Knowledge + Role + Task + Data and provides the necessary context and information for the LLM's analysis.

Knowledge of Campaign Metrics: Evaluation metrics span different facets of an ad campaign, from audience engagement to cost efficiency; they are explained in detail in the prompt and include Reach, Frequency, Results, Cost per Result, Amount Spent, CPM, CTR, CR Link Clicks to LP View, and CR Link Clicks to Result (see Section IX-A). Each metric serves a distinct purpose in evaluating campaign performance and offers valuable insights into audience interaction, cost-effectiveness, and conversion rates.

Role of the LLM: The prompt also defines the specific role that the LLM is supposed to play, namely operate as a surrogate performance marketing manager. The LLM is expected to embody confidence in its analyses, refraining from speculative language such as "could be" or "might be". In this simulated role, the LLM is entrusted with the responsibility of scrutinizing performance metrics and providing well-founded explanations for observed trends. This confidence aligns with the typical expertise and authority of a seasoned performance marketing professional, which improves the resulting analysis.

Task Definition for the LLM: The task assigned to the LLM is explicitly outlined in the prompt. The primary objective is to achieve a lower CPR, and the manager dynamically adjusts the budget to optimize this metric. The task involves not only tracking changes in key metrics such as CPR and Spend but also understanding the trends in secondary metrics such as CTR, CPM, CR, and CR Link Clicks to LP View.

	Original Image	Remove CTA Button	Remove Visual Element	Remove Transparent BG
				
Actual CTR	5.08	3.96	2.12	1.85
Pred Perf.	Above Ave: 0.859	Above Ave: 0.777	Average: 0.448	Below Average: 0.770

Fig. 9. Performance comparison of creatives after object removal.

TABLE V
SAMPLE AD CAMPAIGN ANALYSIS RESULTS BY GPT-4

1) *Key Metrics Change*: Both CPR (Cost Per Result) and SPEND did change over the course of the campaign.
2) *Key Metrics Trend*:

- CPR started at $57.01, decreased to $33.08 at its lowest point, spiked to $208.28, and then decreased again to $27.68 before rising to $115.98. The trend indicates a fluctuation in cost efficiency, with an overall decrease towards the middle of the campaign followed by a substantial increase towards the end.
- SPEND varied significantly, with an initial value of $2565.39, rising to $4664.91, then dropping to $175.18, and finally increasing to $695.91. The campaign budget was dynamically adjusted, leading to these variations.

3) *Secondary Metrics Trend*:

- CTR (Click-Through Rate) fluctuated, starting at 2.68%, peaking at 3.59%, and ending at 2.24%. Changes in CTR indicate varying levels of user engagement with the ads.
- CPM (Cost per 1,000 Impressions) decreased to $9.98, spiked to $14.06, and then dropped to $6.13 before rising again to $11.75. This suggests that ad performance and competition for audience attention were inconsistent.
- CR (Conversion Rate) and CR Link Clicks to LP View showed fluctuations, with a general downward trend towards the end of the campaign. This could indicate a decrease in ad relevance or landing page effectiveness.

4) *Relation between Key and Secondary Metrics*: The increase in CTR typically suggests more effective ad content, which should lower CPR by generating more leads or purchases for the same or less spend. However, the spike in CPM indicates a higher cost to reach the audience, which can increase CPR if the rise in ad costs is not offset by a proportionate increase in purchases. The downward trend in conversion rates, both overall and from link clicks to landing page views, directly contributes to a higher CPR, as fewer users who click the ad proceed to purchase.
5) *Impact of Secondary Metrics on Primary Metrics*: CTR directly influences CPR; a higher CTR means more clicks for the budget, increasing the opportunity for conversions and potentially lowering the CPR if the conversion rate is stable or improves. Conversely, an increase in CPM raises the cost to gain impressions, potentially increasing the CPR if not balanced by a rise in conversion rates. The conversion rate from link clicks to landing page views is pivotal; a decrease suggests that users are less compelled by the landing page to make a purchase, thus increasing CPR. The correlation is clear: while a higher CTR and improved landing page conversion rates contribute to a lower CPR, increases in CPM or declines in conversion rates can negate these benefits, leading to a higher CPR. The data shows fluctuations in CTR and CPM, alongside a general decrease in conversion rates, explaining the observed changes in CPR.
6) *Hypothesis for Changes in Primary Metrics*:

- The initial decrease in CPR can be attributed to the campaign's optimization and improvements in ad targeting, leading to a more engaged audience and more efficient spending.
- The spike in CPR could be due to market dynamics, such as increased competition, leading to higher costs for reaching the audience.
- The subsequent decrease in CPR and then increase suggests that the audience may have experienced ad fatigue, resulting in lower conversion rates and higher costs per purchase.

7) *Analysis of Textual and Visual Data of Ads*:

- The ad's textual content is clear and direct, with a focus on the Keto diet, which is relevant to the target audience. However, the visual content, while appealing, does not directly link to the desired action of taking a quiz or making a purchase.
- To improve the key metrics, I recommend testing variations of the ad copy that directly address the benefits of the Keto diet and include a strong call-to-action. Additionally, using visuals that better represent the quiz or the benefits of following the Keto diet may improve conversion rates.

8) *Recommendation for Next Steps*:

- Monitor the CTR and conversion rates closely and test different ad creatives to find the most engaging content for the audience.
- Analyze the times when CPM spikes to determine if there are patterns related to increased competition and adjust bidding strategies accordingly.
- Review the landing page experience to ensure it is optimized for conversions, with fast load times and clear calls-to-action.
- *Consider retargeting campaigns to engage users who have previously interacted with the ads but have not converted, using tailored messaging based on their interaction stage.*
- Allocate budget dynamically, but with caution during periods of high CPM to avoid inefficiency, focusing more on periods with lower CPM and higher engagement.

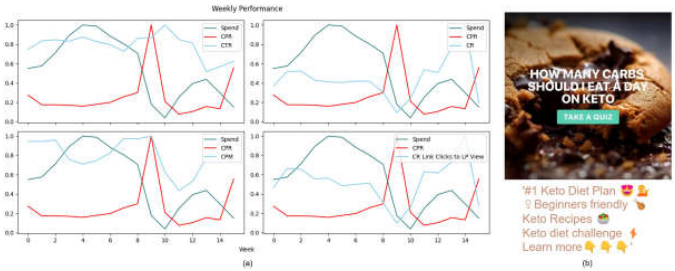


Fig. 10. Data for Section IX-B: (a) weekly performance; (b) a sample creative.

Key Questions for LLM Analysis: To guide the LLM through a systematic analysis, we present in the prompt a set of key questions inspired by the routine thought process of an ad optimization manager. These questions include the following.

1) Did the key metrics (CPR, SPEND) change?
2) If they changed, summarize the trend of the key metrics.
3) What was the trend of secondary metrics (CTR, CPM, CR, CR Link Clicks to LP View) during the same time?
4) Are the changes in key and secondary metrics related?
5) Which secondary metrics influence the key metrics?
6) Provide hypotheses to explain the changes in key metrics.
7) Analyze textual and visual data of ads to explain key metric changes and recommend improvements.
8) Recommend the next steps for campaign performance improvement based on the metrics.

Tabular Data Integration: The prompt is enriched with tabular data that contains both numerical values of metrics and detailed creative features of the ads. This allows the LLM to analyze both quantitative and qualitative aspects, grounding its understanding of the campaign performance and helping generate actionable recommendations for optimization.

Visual Data Integration: In addition to text and tabular data, we included the ads' visuals to enrich contextual information within the prompt. Thus, a multimodal LLM such as GPT-4 (that we used in the experiments) can not only comprehend textual nuances but also mine insights from the visual contents of the campaign, again leading to a more holistic understanding of the relations between creative content and performance.

D. Analysis of the Results

Here, we present the LLM-based analysis for performance metrics evaluation in the specified marketing campaign. The examination primarily focuses on key metrics, such as CPR and Amount Spent, alongside with secondary metrics like CTR, CPM, CR, and CR Link Clicks to LP View. The goal is to uncover insights into the dynamics of the campaign and provide actionable recommendations for improvement. LLM results are shown in Table V. The LLM's outputs demonstrate its proficiency in annotating data and accurately detailing variations in key metrics, as seen in points 1, 2, and 3. This capability is important for understanding recent campaign performances and for summarizing critical data for marketing managers in preparation for reporting meetings. Moreover, the model is quite adept in identifying the root causes of increasing the cost per result metric. It points out the decline in conversion rates and the rise in CPM in points 4 and 5 as contributing factors. In point 6, the LLM goes further, recognizing potential issues such as ad fatigue, ad targeting optimization, and market seasonality. This is especially important because these aspects were not explicitly input in the prompt or data, indicating the model's ability to extract new insights. In a manner similar to the analysis in Section VII, point 7 discusses potential enhancements in visual content and explores the reasons for its current inefficiency. However, point 8 shows a limitation: suggestions provided there are quite generic and may not fully align with the specific needs of the keto diet campaign. These recommendations should be customized to better suit the unique aspects of the campaign. Despite these limitations, the analysis provided by the LLM in this example proves to be a valuable asset for digital marketing professionals, offering quick evaluations and identifying potential areas for optimization across various campaigns. Initial recommendations provided by our platform are especially helpful for early-career marketers and serve well for educational purposes. However, more advanced marketing professionals may require more specific and detailed insights, which we plan to explore in future research.

X. Conclusion

In this work, we have introduced an innovative explainable AI framework *SOMONITOR* that aims to combine human intuition with the efficiency of modern AI to significantly enhance the capabilities of marketers across all stages of the marketing funnel, from strategic planning to content creation and campaign execution. The challenges traditionally faced by marketers, especially the need to manually sift through immense quantities of transient online content, are addressed by *SOMONITOR*'s ability to automate and refine the process of competitor analysis, content research, and strategic branding, which has been proven by a case study in the real-world settings. *SOMONITOR* ranks advertising content by its potential to achieve higher click-through rates via a specially trained model and then employs LLMs to distill core content pillars, which lead to communication themes and targeted customer personas, providing a structured basis for content and campaign planning. By integrating these insights with brand marketing data, *SOMONITOR* not only constructs compelling narratives for new customer personas but also generates detailed content briefs in the form of customer persona stories, empowering marketing teams to produce content and execute campaigns in a guided manner, which has been proven to help in prioritizing different content elements and enhance real-world advertising performnace. We believe that *SOMONITOR* can transform digital marketing operations, enabling marketers to quickly navigate through large datasets and derive actionable insights that boost campaign effectiveness and enhance job satisfaction, redefining traditional marketing approaches [111].

XI. Acknowledgement

This work was funded by the Russian Science Foundation grant №. 22-11-00135, https://rscf.ru/en/project/22-11-00135/

References

[1] S. Makridakis, F. Petropoulos, and Y. Kang, "Large language models: Their success and impact," *Forecasting*, vol. 5, no. 3, pp. 536–549, 2023. [Online]. Available: https://www.mdpi.com/2571-9394/5/3/30

[2] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, and J. Gao, "Large language models: A survey," 2024.

[3] M. Enis and M. Hopkins, "From llm to nmt: Advancing low-resource machine translation with claude," 2024.

[4] M. C. et al., "Evaluating large language models trained on code," 2021.

[5] A. Q. J. et al., "Mistral 7b," 2023.

[6] D. Kevian, U. Syed, X. Guo, A. Havens, G. Dullerud, P. Seiler, L. Qin, and B. Hu, "Capabilities of large language models in control engineering: A benchmark study on gpt-4, claude 3 opus, and gemini 1.0 ultra," 2024.

[7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *CoRR*, vol. abs/1706.03762, 2017. [Online]. Available: http://arxiv.org/abs/1706.03762

[8] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," 2020.

[9] R. Child, S. Gray, A. Radford, and I. Sutskever, "Generating long sequences with sparse transformers," 2019.

[10] S. Wang, B. Z. Li, M. Khabsa, H. Fang, and H. Ma, "Linformer: Self-attention with linear complexity," 2020.

[11] K. Choromanski, V. Likhosherstov, D. Dohan, X. Song, A. Gane, T. Sarlos, P. Hawkins, J. Davis, A. Mohiuddin, L. Kaiser, D. Belanger, L. Colwell, and A. Weller, "Rethinking attention with performers," 2022.

[12] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *First Conference on Language Modeling*, 2024. [Online]. Available: https://openreview.net/forum?id=tEYskw1VY2

[13] H. Qu, L. Ning, R. An, W. Fan, T. Derr, H. Liu, X. Xu, and Q. Li, "A survey of mamba," 2024. [Online]. Available: https://arxiv.org/abs/2408.01129

[14] S. Bansal, S. A, M. P. J, M. S, S. Madisetty, M. Z. U. Rehman, C. S. Raghaw, G. Duggal, and N. Kumar, "A comprehensive survey of mamba architectures for medical image analysis: Classification, segmentation, restoration and beyond," 2024. [Online]. Available: https://arxiv.org/abs/2410.02362

[15] X. Ma, C. Zhou, X. Kong, J. He, L. Gui, G. Neubig, J. May, and L. Zettlemoyer, "Mega: Moving average equipped gated attention," 2023.

[16] A. Orvieto, S. L. Smith, A. Gu, A. Fernando, C. Gulcehre, R. Pascanu, and S. De, "Resurrecting recurrent neural networks for long sequences," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, Eds., vol. 202. PMLR, 23–29 Jul 2023, pp. 26 670–26 698. [Online]. Available: https://proceedings.mlr.press/v202/orvieto23a.html

[17] G. Team, "Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context," 2024.

[18] Anthropic, "Introducing the next generation of claude," https://www.anthropic.com/news/claude-3-family, 2024.

[19] Y. Tang, J. Bi, S. Xu, L. Song, S. Liang, T. Wang, D. Zhang, J. An, J. Lin, R. Zhu, A. Vosoughi, C. Huang, Z. Zhang, F. Zheng, J. Zhang, P. Luo, J. Luo, and C. Xu, "Video understanding with large language models: A survey," 2024.

[20] OpenAI, "Hello gpt-4o," https://openai.com/index/hello-gpt-4o/, 2024.

[21] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 6840–6851.

[22] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations*, 2021. [Online]. Available: https://openreview.net/forum?id=St1giarCHLP

[23] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 10 684–10 695.

[24] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, and T. Salimans, "Imagen video: High definition video generation with diffusion models," 2022.

[25] W. Hong, M. Ding, W. Zheng, X. Liu, and J. Tang, "Cogvideo: Large-scale pretraining for text-to-video generation via transformers," 2022.

[26] Y. Liu, K. Zhang, Y. Li, Z. Yan, C. Gao, R. Chen, Z. Yuan, Y. Huang, H. Sun, J. Gao, L. He, and L. Sun, "Sora: A review on background, technology, limitations, and opportunities of large vision models," 2024.

[27] D. Kocetkov, R. Li, L. B. allal, J. LI, C. Mou, Y. Jernite, M. Mitchell, C. M. Ferrandis, S. Hughes, T. Wolf, D. Bahdanau, L. V. Werra, and H. de Vries, "The stack: 3 TB of permissively licensed source code," *Transactions on Machine Learning Research*, 2023. [Online]. Available: https://openreview.net/forum?id=pxpbTdUEpD

[28] L. Gao, S. Biderman, S. Black, L. Golding, T. Hoppe, C. Foster, J. Phang, H. He, A. Thite, N. Nabeshima, S. Presser, and C. Leahy, "The pile: An 800GB dataset of diverse text for language modeling," *CoRR*, vol. abs/2101.00027, 2021. [Online]. Available: https://arxiv.org/abs/2101.00027

[29] J. M. Patel and J. M. Patel, "Introduction to common crawl datasets," *Getting structured data from the internet: running web crawlers/scrapers on a big data production scale*, pp. 277–324, 2020.

[30] Y. Tokpanov, B. Millidge, P. Glorioso, J. Pilault, A. Ibrahim, J. Whittington, and Q. Anthony, "Zyda: A 1.3t dataset for open language modeling," 2024. [Online]. Available: https://arxiv.org/abs/2406.01981

[31] A. S. Luccioni and J. D. Viviano, "What's in the box? an analysis of undesirable content in the common crawl corpus," *CoRR*, vol. abs/2105.02732, 2021. [Online]. Available: https://arxiv.org/abs/2105.02732

[32] J. W. Rae, S. Borgeaud, T. Cai, K. Millican, J. Hoffmann, H. F. Song, J. Aslanides, S. Henderson, R. Ring, S. Young, E. Rutherford, T. Hennigan, J. Menick, A. Cassirer, R. Powell, G. van den Driessche, L. A. Hendricks, M. Rauh, P. Huang, A. Glaese, J. Welbl, S. Dathathri, S. Huang, J. Uesato, J. Mellor, I. Higgins, A. Creswell, N. McAleese, A. Wu, E. Elsen, S. M. Jayakumar, E. Buchatskaya, D. Budden, E. Sutherland, K. Simonyan, M. Paganini, L. Sifre, L. Martens, X. L. Li, A. Kuncoro, A. Nematzadeh, E. Gribovskaya, D. Donato, A. Lazaridou, A. Mensch, J. Lespiau, M. Tsimpoukelli, N. Grigorev, D. Fritz, T. Sottiaux, M. Pajarskas, T. Pohlen, Z. Gong, D. Toyama, C. de Masson d'Autume, Y. Li, T. Terzi, V. Mikulik, I. Babuschkin, A. Clark, D. de Las Casas, A. Guy, C. Jones, J. Bradbury, M. J. Johnson, B. A. Hechtman, L. Weidinger, I. Gabriel, W. Isaac, E. Lockhart, S. Osindero, L. Rimell, C. Dyer, O. Vinyals, K. Ayoub, J. Stanway, L. Bennett, D. Hassabis, K. Kavukcuoglu, and G. Irving, "Scaling language models: Methods, analysis & insights from training gopher," *CoRR*, vol. abs/2112.11446, 2021. [Online]. Available: https://arxiv.org/abs/2112.11446

[33] Y. Elazar, A. Bhagia, I. H. Magnusson, A. Ravichander, D. Schwenk, A. Suhr, E. P. Walsh, D. Groeneveld, L. Soldaini, S. Singh, H. Hajishirzi, N. A. Smith, and J. Dodge, "What's in my big data?" in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=RvfPnOkPV4

[34] K. Lee, D. Ippolito, A. Nystrom, C. Zhang, D. Eck, C. Callison-Burch, and N. Carlini, "Deduplicating training data makes language models better," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, S. Muresan, P. Nakov, and A. Villavicencio, Eds. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 8424–8445. [Online]. Available: https://aclanthology.org/2022.acl-long.577

[35] S. Min, S. Gururangan, E. Wallace, W. Shi, H. Hajishirzi, N. A. Smith, and L. Zettlemoyer, "SILO language models: Isolating legal risk in a nonparametric datastore," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=ruk0nyQPec

[36] Z. Yu, Y. Wu, N. Zhang, C. Wang, Y. Vorobeychik, and C. Xiao, "CodeIPPrompt: Intellectual property infringement assessment of code language models," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research,

A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, Eds., vol. 202. PMLR, 23–29 Jul 2023, pp. 40 373–40 389. [Online]. Available: https://proceedings.mlr.press/v202/yu23g.html

[37] P. Samuelson, "Generative ai meets copyright," *Science*, vol. 381, no. 6654, pp. 158–161, 2023. [Online]. Available: https://www.science.org/doi/abs/10.1126/science.adi0656

[38] S. Longpre, L. Hou, T. Vu, A. Webson, H. W. Chung, Y. Tay, D. Zhou, Q. V. Le, B. Zoph, J. Wei, and A. Roberts, "The flan collection: designing data and methods for effective instruction tuning," in *Proceedings of the 40th International Conference on Machine Learning*, ser. ICML'23. JMLR.org, 2023.

[39] M. Conover, M. Hayes, A. Mathur, J. Xie, J. Wan, S. Shah, A. Ghodsi, P. Wendell, M. Zaharia, and R. Xin. (2023) Free dolly: Introducing the world's first truly open instruction-tuned llm. [Online]. Available: https://www.databricks.com/blog/2023/04/12/dolly-first-open-commercially-viable-instruction-tuned-llm

[40] V. Sanh, A. Webson, C. Raffel, S. Bach, L. Sutawika, Z. Alyafeai, A. Chaffin, A. Stiegler, A. Raja, M. Dey, M. S. Bari, C. Xu, U. Thakker, S. S. Sharma, E. Szczechla, T. Kim, G. Chhablani, N. Nayak, D. Datta, J. Chang, M. T.-J. Jiang, H. Wang, M. Manica, S. Shen, Z. X. Yong, H. Pandey, R. Bawden, T. Wang, T. Neeraj, J. Rozen, A. Sharma, A. Santilli, T. Fevry, J. A. Fries, R. Teehan, T. L. Scao, S. Biderman, L. Gao, T. Wolf, and A. M. Rush, "Multitask prompted training enables zero-shot task generalization," in *International Conference on Learning Representations*, 2022. [Online]. Available: https://openreview.net/forum?id=9Vrb9D0WI4

[41] Y. Liu, S. Tao, X. Zhao, M. Zhu, W. Ma, J. Zhu, C. Su, Y. Hou, M. Zhang, M. Zhang, H. Ma, L. Zhang, H. Yang, and Y. Jiang, "Coachlm: Automatic instruction revisions improve the data quality in llm instruction tuning," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 5184–5197.

[42] W. Ni, K. Zhang, X. Miao, X. Zhao, Y. Wu, and J. Yin, "Iterclean: An iterative data cleaning framework with large language models," in *Proceedings of the ACM Turing Award Celebration Conference - China 2024*, ser. ACM-TURC '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 100–105. [Online]. Available: https://doi.org/10.1145/3674399.3674436

[43] Y. Wang, S. Mishra, P. Alipoormolabashi, Y. Kordi, A. Mirzaei, A. Arunkumar, A. Ashok, A. S. Dhanasekaran, A. Naik, D. Stap, E. Pathak, G. Karamanolakis, H. G. Lai, I. Purohit, I. Mondal, J. Anderson, K. Kuznia, K. Doshi, M. Patel, K. K. Pal, M. Moradshahi, M. Parmar, M. Purohit, N. Varshney, P. R. Kaza, P. Verma, R. S. Puri, R. Karia, S. K. Sampat, S. Doshi, S. Mishra, S. Reddy, S. Patro, T. Dixit, X. Shen, C. Baral, Y. Choi, N. A. Smith, H. Hajishirzi, and D. Khashabi, "Super-naturalinstructions: Generalization via declarative instructions on 1600+ nlp tasks," 2022. [Online]. Available: https://arxiv.org/abs/2204.07705

[44] O. Honovich, T. Scialom, O. Levy, and T. Schick, "Unnatural instructions: Tuning language models with (almost) no human labor," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 14 409–14 428. [Online]. Available: https://aclanthology.org/2023.acl-long.806

[45] T. Li, Q. Liu, T. Pang, C. Du, Q. Guo, Y. Liu, and M. Lin, "Purifying large language models by ensembling a small language model," 2024. [Online]. Available: https://arxiv.org/abs/2402.14845

[46] S. Zhang, L. Dong, X. Li, S. Zhang, X. Sun, S. Wang, J. Li, R. Hu, T. Zhang, F. Wu, and G. Wang, "Instruction tuning for large language models: A survey," 2024. [Online]. Available: https://arxiv.org/abs/2308.10792

[47] H. Zhang, Y. Dong, C. Xiao, and M. Oyamada, "Jellyfish: Instruction-tuning local large language models for data preprocessing," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 8754–8782. [Online]. Available: https://aclanthology.org/2024.emnlp-main.497

[48] F. Biester, M. Abdelaal, and D. Del Gaudio, "Llmclean: Context-aware tabular data cleaning via llm-generated ofds," in *New Trends in Database and Information Systems*, J. Tekli, J. Gamper, R. Chbeir, Y. Manolopoulos, S. Sassi, M. Ivanovic, G. Vargas-Solar, and E. Zumpano, Eds. Cham: Springer Nature Switzerland, 2025, pp. 68–78.

[49] N. Hollmann, S. Müller, and F. Hutter, "Large language models for automated data science: Introducing CAAFE for context-aware automated feature engineering," in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. [Online]. Available: https://openreview.net/forum?id=9WSxQZ9mG7

[50] G. Badaro, M. Saeed, and P. Papotti, "Transformers for tabular data representation: A survey of models and applications," *Transactions of the Association for Computational Linguistics*, vol. 11, pp. 227–249, 2023. [Online]. Available: https://aclanthology.org/2023.tacl-1.14

[51] Y. Sui, M. Zhou, M. Zhou, S. Han, and D. Zhang, "Table meets llm: Can large language models understand structured table data? a benchmark and empirical study," in *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, ser. WSDM '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 645–654. [Online]. Available: https://doi.org/10.1145/3616855.3635752

[52] V. Borisov, K. Sessler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=cEygmQNOeI

[53] S. Arora, B. Yang, S. Eyuboglu, A. Narayan, A. Hojel, I. Trummer, and C. Ré, "Language models enable simple systems for generating structured views of heterogeneous data lakes," *Proc. VLDB Endow.*, vol. 17, no. 2, p. 92–105, Oct. 2023. [Online]. Available: https://doi.org/10.14778/3626292.3626294

[54] Y. Lou, C. Lei, X. Qin, Z. Wang, C. Faloutsos, R. Anubhai, and H. Rangwala, "Datalore: Can a large language model find all lost scrolls in a data repository?" in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 5170–5176.

[55] M. Ganti, L. Orr, and S. Wu, "Evaluating text-to-sql model failures on real-world data," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 1–1.

[56] T. Ren, Y. Fan, Z. He, R. Huang, J. Dai, C. Huang, Y. Jing, K. Zhang, Y. Yang, and X. S. Wang, "Purple: Making a large language model a better sql writer," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 15–28.

[57] Y. Fan, Z. He, T. Ren, D. Guo, L. Chen, R. Zhu, G. Chen, Y. Jing, K. Zhang, and X. S. Wang, "Gar: A generate-and-rank approach for natural language to sql translation," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 110–122.

[58] Y. Fan, T. Ren, Z. He, X. Wang, Y. Zhang, and X. Li, "Gensql: A generative natural language interface to database systems," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 3603–3606.

[59] E. Veltri, G. Badaro, M. Saeed, and P. Papotti, "Data ambiguity profiling for the generation of training examples," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 450–463.

[60] J.-P. Zhu, P. Cai, B. Niu, Z. Ni, K. Xu, J. Huang, J. Wan, S. Ma, B. Wang, D. Zhang, L. Tang, and Q. Liu, "Chat2query: A zero-shot automatic exploratory data analysis system with large language models," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 5429–5432.

[61] Y. Peng, S. Lin, Q. Chen, S. Wang, L. Xu, X. Ren, Y. Li, and J. Xu, "Chatgraph: Chat with your graphs," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 5445–5448.

[62] Z. Qin, C. Luo, Z. Wang, H. Jiang, and Y. Sun, "Relational database augmented large language model," 2024. [Online]. Available: https://arxiv.org/abs/2407.15071

[63] G. Li, X. Zhou, and X. Zhao, "Llm for data management," *Proc. VLDB Endow.*, vol. 17, no. 12, p. 4213–4216, Nov. 2024. [Online]. Available: https://doi.org/10.14778/3685800.3685838

[64] X. Zhou, Z. Sun, and G. Li, "Db-gpt: Large language model meets database," *Data Science and Engineering*, vol. 9, pp. 1–10, 01 2024.

[65] Z. Zhao, W. Fan, J. Li, Y. Liu, X. Mei, Y. Wang, Z. Wen, F. Wang, X. Zhao, J. Tang, and Q. Li, "Recommender Systems in the Era of Large Language Models (LLMs)," *IEEE Transactions on Knowledge & Data Engineering*, vol. 36, no. 11, pp. 6889–6907, Nov. 2024. [Online]. Available: https://doi.ieeecomputersociety.org/10.1109/TKDE.2024.3392335

[66] J. Liu, C. Liu, P. Zhou, R. Lv, K. Zhou, and Y. Zhang, "Is chatgpt a good recommender? a preliminary study," 2023. [Online]. Available: https://arxiv.org/abs/2304.10149

[67] J. Chen, L. Ma, X. Li, N. Thakurdesai, J. Xu, J. H. D. Cho, K. Nag, E. Korpeoglu, S. Kumar, and K. Achan, "Knowledge graph completion models are few-shot learners: An empirical study of relation labeling in e-commerce with llms," 2023. [Online]. Available: https://arxiv.org/abs/2305.09858

[68] X. Wang, X. Tang, X. Zhao, J. Wang, and J.-R. Wen, "Rethinking the evaluation for conversational recommendation in the era of large language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, H. Bouamor, J. Pino, and K. Bali, Eds. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 10 052–10 065. [Online]. Available: https://aclanthology.org/2023.emnlp-main.621

[69] Z. He, Z. Xie, R. Jha, H. Steck, D. Liang, Y. Feng, B. P. Majumder, N. Kallus, and J. Mcauley, "Large language models as zero-shot conversational recommenders," in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, ser. CIKM '23. ACM, Oct. 2023, p. 720–730. [Online]. Available: http://dx.doi.org/10.1145/3583780.3614949

[70] Y. Gao, T. Sheng, Y. Xiang, Y. Xiong, H. Wang, and J. Zhang, "Chat-rec: Towards interactive and explainable llms-augmented recommender system," 2023. [Online]. Available: https://arxiv.org/abs/2303.14524

[71] S. Geng, S. Liu, Z. Fu, Y. Ge, and Y. Zhang, "Recommendation as language processing (rlp): A unified pretrain, personalized prompt & predict paradigm (p5)," in *Proceedings of the 16th ACM Conference on Recommender Systems*, ser. RecSys '22. New York, NY, USA: Association for Computing Machinery, 2022, p. 299–315. [Online]. Available: https://doi.org/10.1145/3523227.3546767

[72] S. Xu, W. Hua, and Y. Zhang, "Openp5: An open-source platform for developing, training, and evaluating llm-based recommender systems," 2024. [Online]. Available: https://arxiv.org/abs/2306.11134

[73] Z. Cui, J. Ma, C. Zhou, J. Zhou, and H. Yang, "M6-rec: Generative pretrained language models are open-ended recommender systems," 2022. [Online]. Available: https://arxiv.org/abs/2205.08084

[74] H. Zhu, H. Ge, X. Gu, P. Zhao, and D. L. Lee, "Influential recommender system," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 1406–1419.

[75] J. Zhang, R. Xie, Y. Hou, W. X. Zhao, L. Lin, and J.-R. Wen, "Recommendation as instruction following: A large language model empowered recommendation approach," 2023. [Online]. Available: https://arxiv.org/abs/2305.07001

[76] L. Friedman, S. Ahuja, D. Allen, Z. Tan, H. Sidahmed, C. Long, J. Xie, G. Schubiner, A. Patel, H. Lara, B. Chu, Z. Chen, and M. Tiwari, "Leveraging large language models in conversational recommender systems," 2023. [Online]. Available: https://arxiv.org/abs/2305.07961

[77] B. Zheng, Y. Hou, H. Lu, Y. Chen, W. X. Zhao, M. Chen, and J.-R. Wen, "Adapting large language models by integrating collaborative semantics for recommendation," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 1435–1448.

[78] K. Bao, J. Zhang, Y. Zhang, W. Wang, F. Feng, and X. He, "Tallrec: An effective and efficient tuning framework to align large language model with recommendation," in *Proceedings of the 17th ACM Conference on Recommender Systems*, ser. RecSys '23. ACM, Sep. 2023, p. 1007–1014. [Online]. Available: http://dx.doi.org/10.1145/3604915.3608857

[79] L. Wu, Z. Qiu, Z. Zheng, H. Zhu, and E. Chen, "Exploring large language model for graph data understanding in online job recommendations," 2023. [Online]. Available: https://arxiv.org/abs/2307.05722

[80] A.-T. Kuo, H. Chen, and W.-S. Ku, "Bert-trip: Effective and scalable trip representation using attentive contrast learning," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 612–623.

[81] L. Zhang, X. Zhou, Z. Zeng, and Z. Shen, "Are id embeddings necessary? whitening pre-trained text embeddings for effective sequential recommendation," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 530–543.

[82] Y. Hou, Z. He, J. McAuley, and W. X. Zhao, "Learning vector-quantized item representation for transferable sequential recommenders," in *Proceedings of the ACM Web Conference 2023*, ser. WWW '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 1162–1171. [Online]. Available: https://doi.org/10.1145/3543507.3583434

[83] Z. Yuan, F. Yuan, Y. Song, Y. Li, J. Fu, F. Yang, Y. Pan, and Y. Ni, "Where to go next for recommender systems? id- vs. modality-based recommender models revisited," in *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 2639–2649. [Online]. Available: https://doi.org/10.1145/3539618.3591932

[84] Y. Hou, S. Mu, W. X. Zhao, Y. Li, B. Ding, and J.-R. Wen, "Towards universal sequence representation learning for recommender systems," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ser. KDD '22. New York, NY, USA: Association for Computing Machinery, 2022, p. 585–593. [Online]. Available: https://doi.org/10.1145/3534678.3539381

[85] J. Wang, F. Yuan, M. Cheng, J. M. Jose, C. Yu, B. Kong, X. He, Z. Wang, B. Hu, and Z. Li, "Transrec: Learning transferable recommendation from mixture-of-modality feedback," 2022. [Online]. Available: https://arxiv.org/abs/2206.06190

[86] Y. Li, H. Du, Y. Ni, P. Zhao, Q. Guo, F. Yuan, and X. Zhou, "Multi-modality is all you need for transferable recommender systems," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 5008–5021.

[87] L. Guo, C. Wang, X. Wang, L. Zhu, and H. Yin, "Automated prompting for non-overlapping cross-domain sequential recommendation," 2023. [Online]. Available: https://arxiv.org/abs/2304.04218

[88] Y. Zhang, Y. Li, L. Cui, D. Cai, L. Liu, T. Fu, X. Huang, E. Zhao, Y. Zhang, Y. Chen *et al.*, "Siren's song in the ai ocean: a survey on hallucination in large language models," *arXiv preprint arXiv:2309.01219*, 2023.

[89] Y. Chen, Q. Fu, Y. Yuan, Z. Wen, G. Fan, D. Liu, D. Zhang, Z. Li, and Y. Xiao, "Hallucination detection: Robustly discerning reliable answers in large language models," in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 2023, pp. 245–255.

[90] X. He, J. Pan, O. Jin, T. Xu, B. Liu, T. Xu, Y. Shi, A. Atallah, R. Herbrich, S. Bowers, and J. Q. Candela, "Practical lessons from predicting clicks on ads at facebook," in *Proc. 8th International Workshop on Data Mining for Online Advertising*, ser. ADKDD'14. ACM, 2014, p. 1–9.

[91] H. B. McMahan, G. Holt, D. Sculley, M. Young, D. Ebner, J. Grady, L. Nie, T. Phillips, E. Davydov, D. Golovin, S. Chikkerur, D. Liu, M. Wattenberg, A. M. Hrafnkelsson, T. Boulos, and J. Kubica, "Ad click prediction: A view from the trenches," in *Proc. 19th ACM SIGKDD*, ser. KDD '13. ACM, 2013, p. 1222–1230.

[92] W. Ouyang, X. Zhang, S. Ren, C. Qi, Z. Liu, and Y. Du, "Representation learning-assisted click-through rate prediction," in *Proc. 28th IJCAI*, 7 2019, pp. 4561–4567.

[93] T. Ge, H. Liu, P. Yi, S. Huang, Z. Zhang, X. Zhu, Y. Zhang, K. Gai, L. Zhao, G. Zhou, K. Chen, S. Liu, H. Yi, Z. Hu, B. Liu, and P. Sun, "Image matters: Visually modeling user behaviors using advanced model server," 10 2018, pp. 2087–2095.

[94] S. Feizi, M. Hajiaghayi, K. Rezaei, and S. Shin, "Online advertisements with llms: Opportunities and challenges," 2024.

[95] E. Thackray and M. Northin, "Leveraging large language models in advertising," WPP Mindshare, https://www.wpp.com/wpp-iq/2023/11/leveraging-large-language-models-in-advertising, 2023.

[96] P. Keni and M. ElBermawy, "Marketing generative ai report: A comprehensive evaluation of leading llms," NoGood, https://nogood.io/2024/03/13/generative-ai-in-marketing/, 2024.

[97] Q. Yang, M. Ongpin, S. Nikolenko, A. Huang, and A. Farseev, "Against opacity: Explainable ai and large language models for effective digital advertising," in *Proceedings of the 31st ACM International Conference on Multimedia*, ser. MM '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 9299–9305. [Online]. Available: https://doi.org/10.1145/3581783.3612817

[98] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. Christiano, J. Leike, and R. Lowe, "Training language models to follow instructions with human feedback," 2022.

[99] OpenAI, "Gpt-4 technical report," 2023.

[100] S. Koltcov, O. Koltsova, and S. Nikolenko, "Latent dirichlet allocation: Stability and applications to studies of user-generated content," in *Proceedings of the 2014 ACM Conference on Web Science*, ser. WebSci '14. New York, NY, USA: ACM, 2014, p. 161–165. [Online]. Available: https://doi.org/10.1145/2615569.2615680

[101] S. Nikolenko, "Svd-lda: Topic modeling for full-text recommender systems," in *Advances in Artificial Intelligence and Its Applications*, O. Pichardo Lagunas, O. Herrera Alcántara, and G. Arroyo Figueroa, Eds. Cham: Springer International Publishing, 2015, pp. 67–79.

[102] E. Tutubalina and S. I. Nikolenko, "Exploring convolutional neural networks and topic models for user profiling from drug reviews," *Multimedia Tools and Applications*, vol. 77, no. 4, pp. 4791–4809, 2018.

[103] K. Buraya, A. Farseev, and A. Filchenkov, "Multi-view personality profiling based on longitudinal data," *Lecture Notes in Computer Science*, vol. 11018, pp. 15–27, 2018.

[104] E. Tutubalina and S. I. Nikolenko, "Demographic prediction based on user reviews about medications," *Computación y Sistemas*, vol. 21, no. 2, pp. 227–241, 2017.

[105] A. Savchenko, A. Alekseev, S. Kwon, E. Tutubalina, E. Myasnikov, and S. Nikolenko, "Ad lingua: Text classification improves symbolism prediction in image advertisements," in *Proceedings of the 28th International Conference on Computational Linguistics*. Barcelona, Spain (Online): International Committee on Computational Linguistics, 12 2020, pp. 1886–1892. [Online]. Available: https://www.aclweb.org/anthology/2020.coling-main.171

[106] A. Malik, "Persona based marketing strategies: creation of personas through data analytics," Master's thesis, A. Malik, 2019.

[107] D. Pelleg, A. W. Moore *et al.*, "X-means: Extending k-means with efficient estimation of the number of clusters." in *Icml*, vol. 1. Stanford, CA, 2000, pp. 727–734.

[108] A. Neelakantan, T. Xu, R. Puri, A. Radford, J. M. Han, J. Tworek, Q. Yuan, N. Tezak, J. W. Kim, C. Hallacy *et al.*, "Text and code embeddings by contrastive pre-training," *arXiv preprint arXiv:2201.10005*, 2022.

[109] Z. Abrams and E. Vee, "Personalized ad delivery when ads fatigue: An approximation algorithm," in *International Workshop on Web and Internet Economics*. Springer, 2007, pp. 535–540.

[110] G. Aggarwal, S. Muthukrishnan, D. Pál, and M. Pál, "General auction mechanism for search advertising," in *Proceedings of the 18th international conference on World wide web*, 2009, pp. 241–250.

[111] A. Farseev, "Redemptive marketing: The path to resolving transparency and performance dilemmas in digital advertising," *Forbes.com*, 2024.